

Syed Ali Abbas Abedi

Research Fellow | NUS Institute for Functional Intelligent Materials

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<https://abedisyedaliabbas.github.io/molecular-design-lab/>

Computational chemistry, excited-state photophysics, molecular design, and delta machine learning. Current work: NRF Materials Data Foundry project, NUS I-FIM.

Research experience

2026-08 - Present

Research Fellow

National University of Singapore, Institute for Functional Intelligent Materials (I-FIM), Singapore

Working on delta machine learning within the NRF Materials Data Foundry project at NUS I-FIM.

2026-01 - 2026-08

Research Fellow

Nanyang Technological University (NTU), School of Chemistry, Chemical Engineering and Biotechnology, Singapore

Investigated excited-state pathways and molecular structure-property relationships, combining quantum chemical calculations with computational workflows and collaboration with experimental teams.

2024-09 - 2026-01

Research Fellow (Postdoctoral)

Singapore University of Technology and Design (SUTD), Singapore

Conducted computational research on organic fluorophores, photostability, and photoinduced processes. Developed workflows for quantum chemical calculations and analysis.

2023-04 - 2023-07

Visiting Researcher

University College London (UCL) & Queen Mary University of London, UK

Worked on excited-state decay pathways and photochemical simulations, including multireference electronic-structure methods and nonadiabatic dynamics.

2020-08 - 2024-06

Graduate Research Assistant

Singapore University of Technology and Design (SUTD), Singapore

Investigated the computational design and photophysical mechanisms of fluorophores, focusing on excited-state conformational dynamics under the supervision of Prof. Xiaogang Liu.

Education

2020 - 2024

PhD in Science, Mathematics & Technology

Singapore University of Technology and Design, Singapore

Defence ("viva"): 04/06/2024 - Awarded: 30/06/2024. Thesis: "Computational Design and Mechanistic Investigation of Fluorophores Based on Excited-State Conformational Dynamics"

2017 - 2019

MS in Mathematics

Lahore University of Management Sciences, Pakistan

Awarded: 30/06/2019. Thesis: "Predator-Prey Model with Age Structures"

2011 - 2016

BSc in Chemical Engineering and Technology

University of Engineering & Technology, Lahore, Pakistan

Awarded: 31/01/2016

Methods and technical skills

Computational Chemistry & Photophysics

Quantum Chemistry; DFT, TD-DFT; CASSCF/CASPT2; Non-Adiabatic Dynamics; MD Simulations; Gaussian, ORCA, OpenMolcas, Avogadro, GAMESS

Data Science & Machine Learning (AI-for-X)

Python (Pandas, NumPy, Matplotlib); Scikit-learn; PyTorch; Data Curation; Feature Engineering; Regression

Programming & Data Processing

Python, Bash Scripting; High-Performance Computing (HPC); PBS Workload Managers; Git/GitHub

Molecular Design & Modeling

Fluorophore Design; Excited-State Dynamics; Structure-Property Relationship Mapping; TICT/PET/ESIPT Mechanism Elucidation

Selected publications

2026 | Nature Chemistry

A general strategy towards multicolour fluorogenic peptides for wash-free bioassays

<https://doi.org/10.1038/s41557-026-02228-0>

2026 | Chemical Society Reviews

Mechanistic insights into the design of fluorogenic molecules for wash-free biological applications

<https://doi.org/10.1039/D5CS01519G>

2025 | Nature Methods

Super-photostable organic dye for long-term live-cell single-protein imaging

<https://doi.org/10.1038/s41592-024-02584-0>

2021 | The Journal of Physical Chemistry A

Restriction of twisted intramolecular charge transfer enables aggregation-induced emission

<https://doi.org/10.1021/acs.jpca.1c06263>

Awards and support

2024 | Awarded a £800 bursary

EPSRC Catalysis Hub and the RSC Summer School in Catalysis

2023 | \$250 APC Exhibitor Prize

12th Asian Photochemistry Conference

2023 | Emerging Area in Photochemistry research grant

12th Asian Photochemistry Conference (\$1,000 for innovative research)

2022 | £500 RSC Travel Grant for Researchers

Royal Society of Chemistry

2022 | Granted \$900 funding

American Institute of Chemical Engineers (AIChE) for attending the Quantum Computing Workshop at Denmark Technical Institute (DTU)

2020-2024 | PhD SUTD Fellowship

Scholarship covering doctoral studies

Selected presentations and meetings

Participated @ Global Young Scientists Summit in Singapore (2025).

Invited speaker @ Interdisciplinary Medicine Young Scholars Forum on Biomedical Applications of Molecular Probes, Singapore (2024).

Participated @ 12th Singapore International Chemistry Conference (2024).

Poster @ Catalysis Fundamentals and Practice Summer School, Liverpool, UK (2024).

Oral speaker @ The 5th International Conference on Fluorescent Biomolecules, Hong Kong (2024).